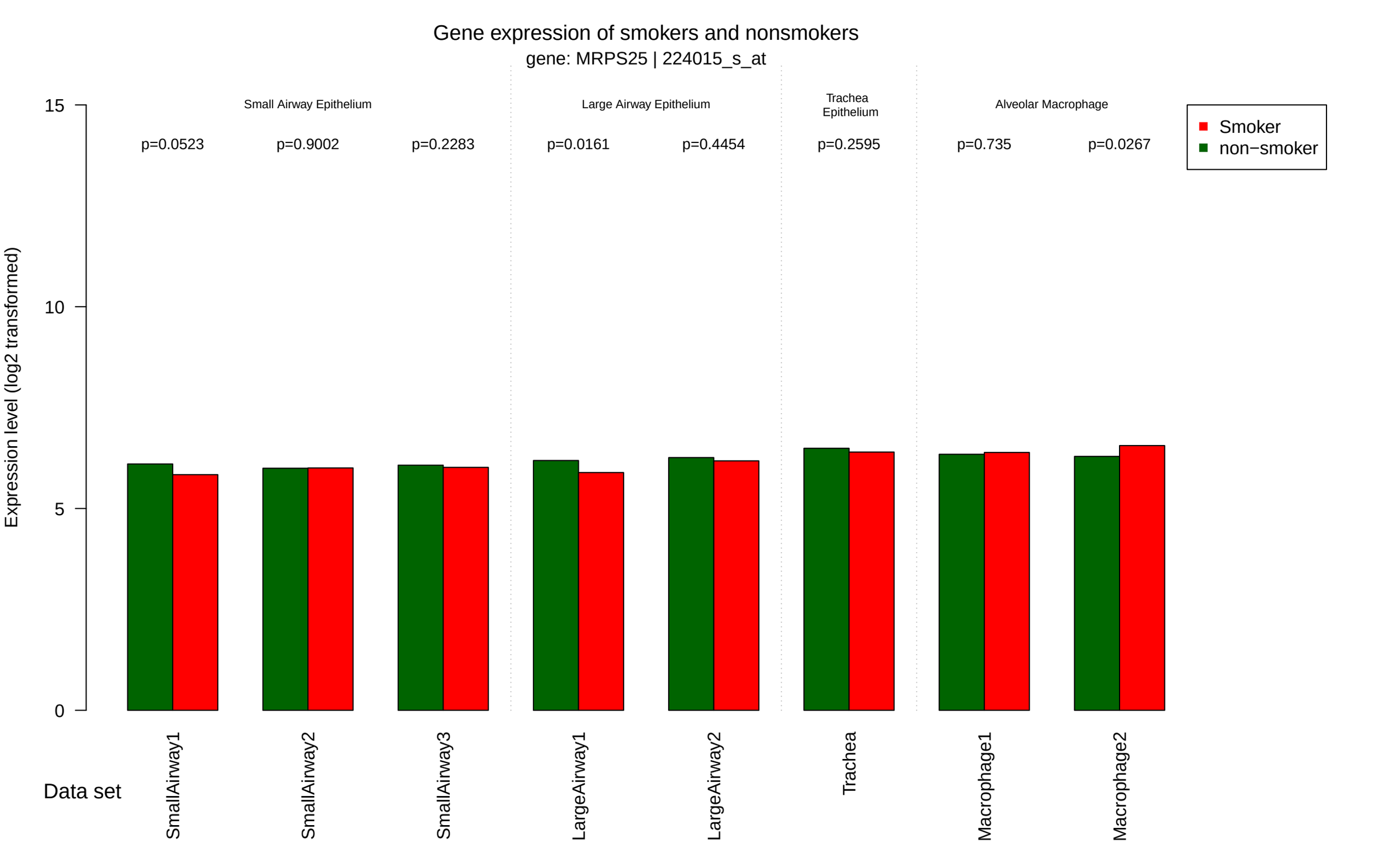
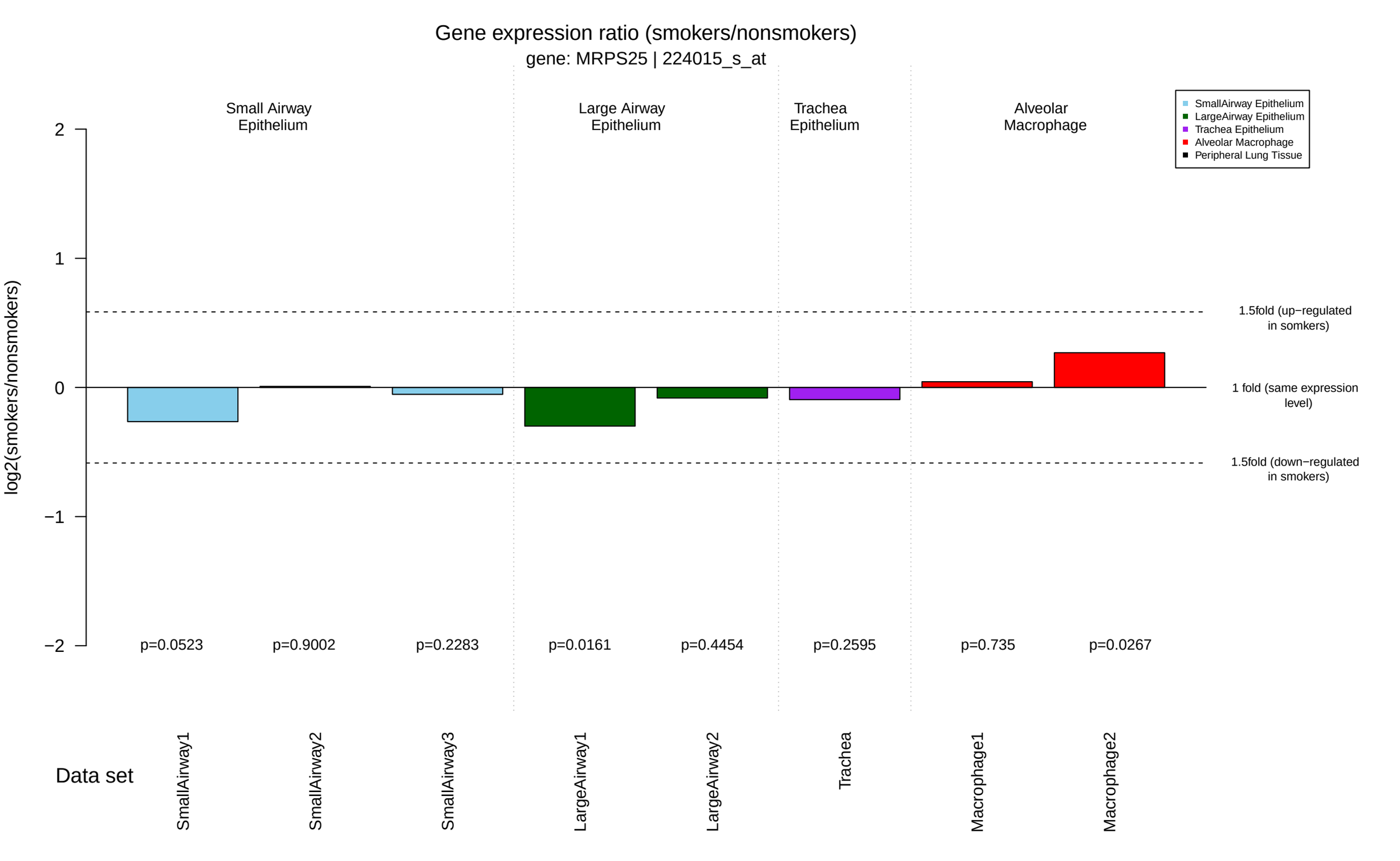
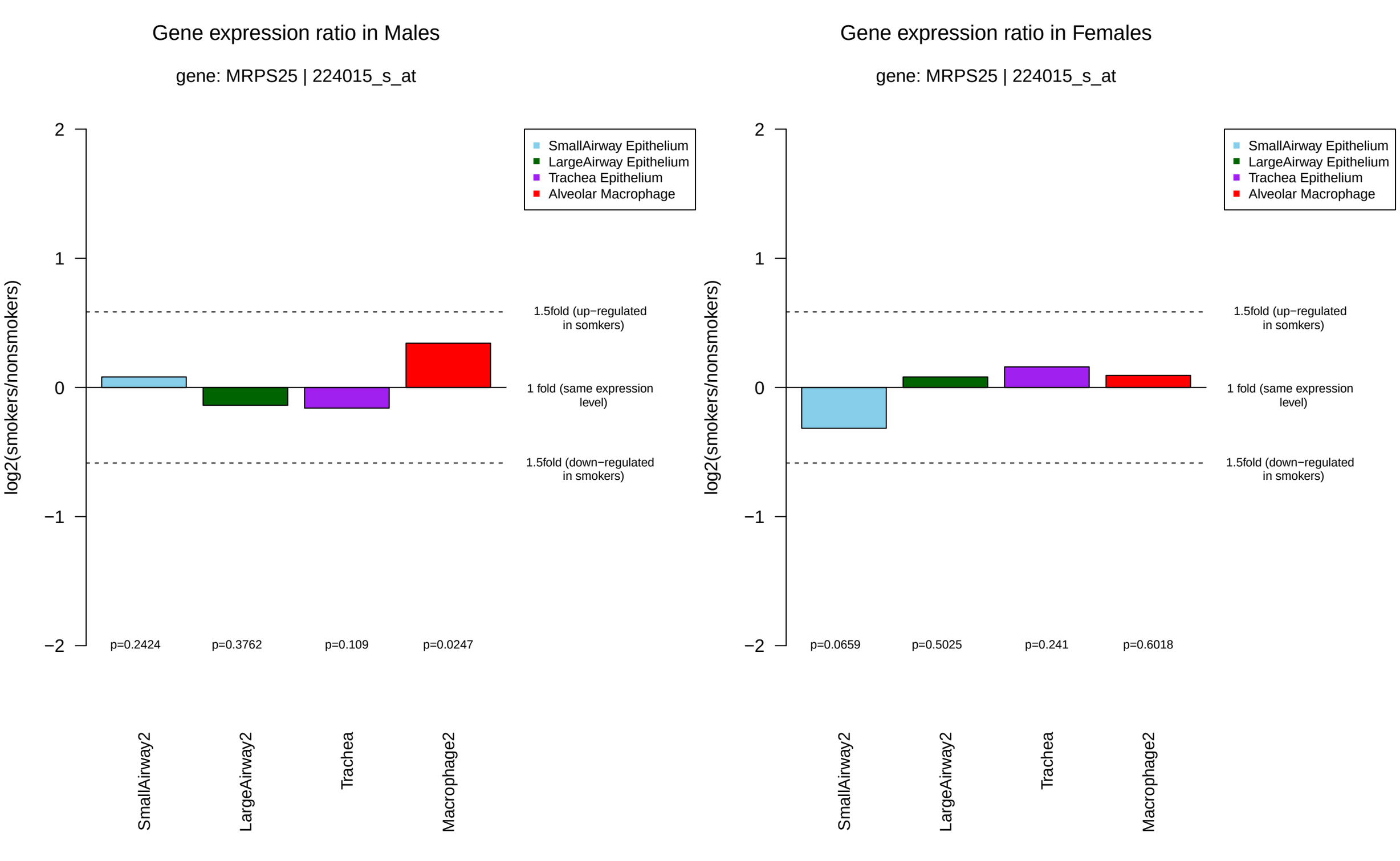
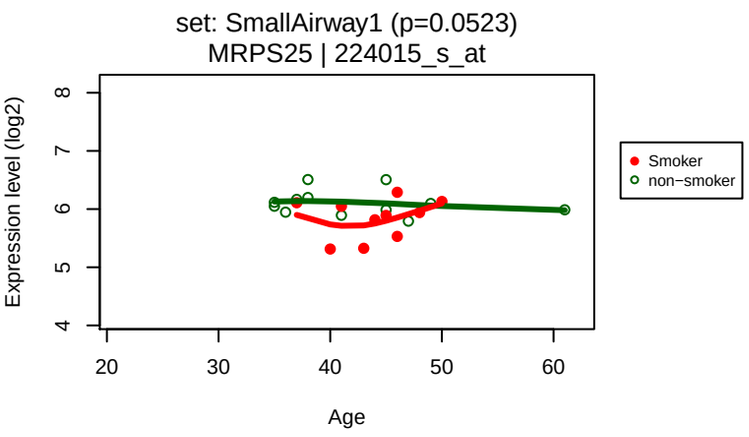
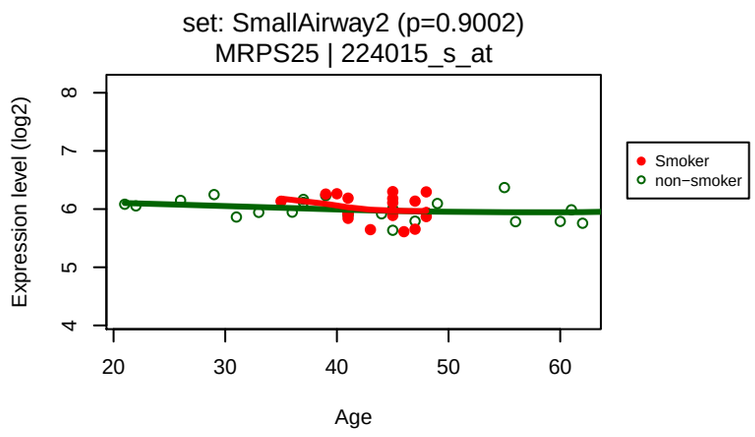
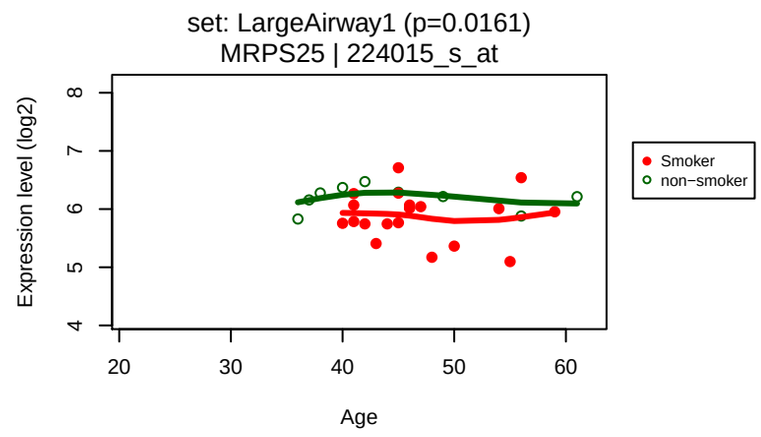

Gene expression of smokers and nonsmokers
gene: MRPS25 | 224015_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0523
p=0.9002
p=0.2283
p=0.0161
p=0.4454
p=0.2595
p=0.735
p=0.0267
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

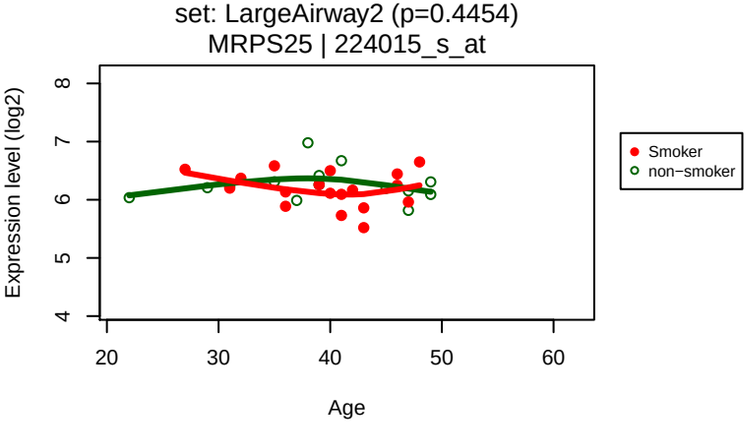

Gene expression ratio (smokers/nonsmokers)
gene: MRPS25 | 224015_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0523
p=0.9002
p=0.2283
p=0.0161
p=0.4454
p=0.2595
p=0.735
p=0.0267
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

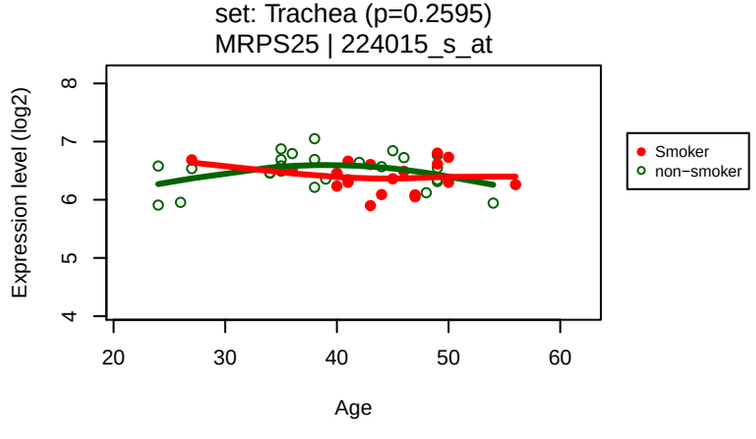

Gene expression ratio in Males
gene: MRPS25 | 224015_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2424
p=0.3762
p=0.109
p=0.0247
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MRPS25 | 224015_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0659
p=0.5025
p=0.241
p=0.6018
SmallAirway2
LargeAirway2
Trachea
Macrophage2

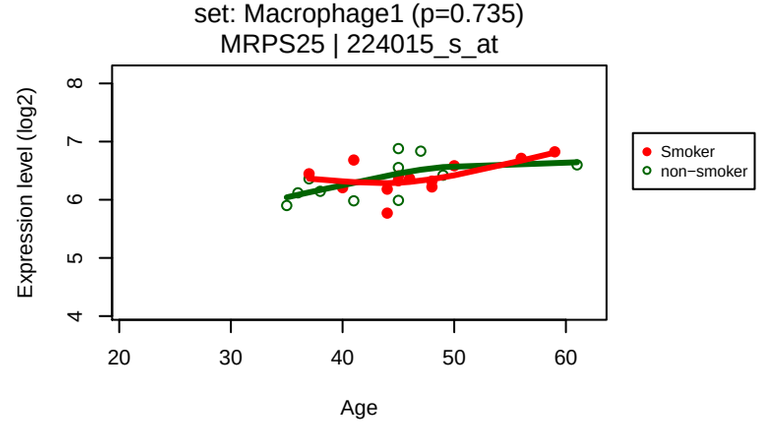

set: SmallAirway1 (p=0.0523)
MRPS25 | 224015_s_at
Expression level (log2)
Age
Smoker
non−smoker

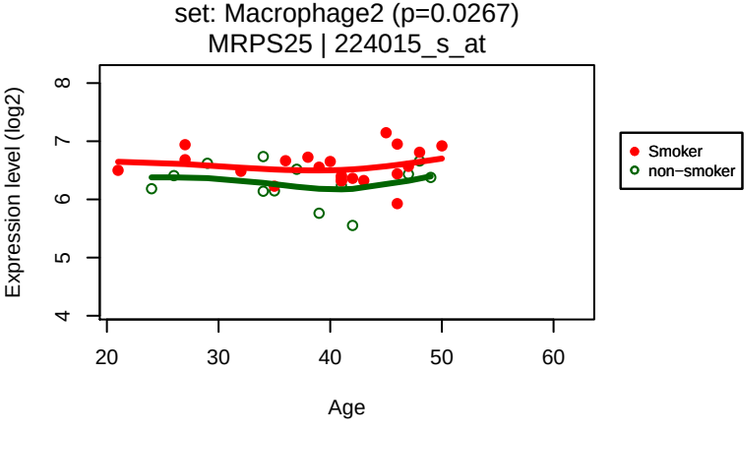

set: SmallAirway2 (p=0.9002)
MRPS25 | 224015_s_at
Expression level (log2)
Age
Smoker
non−smoker

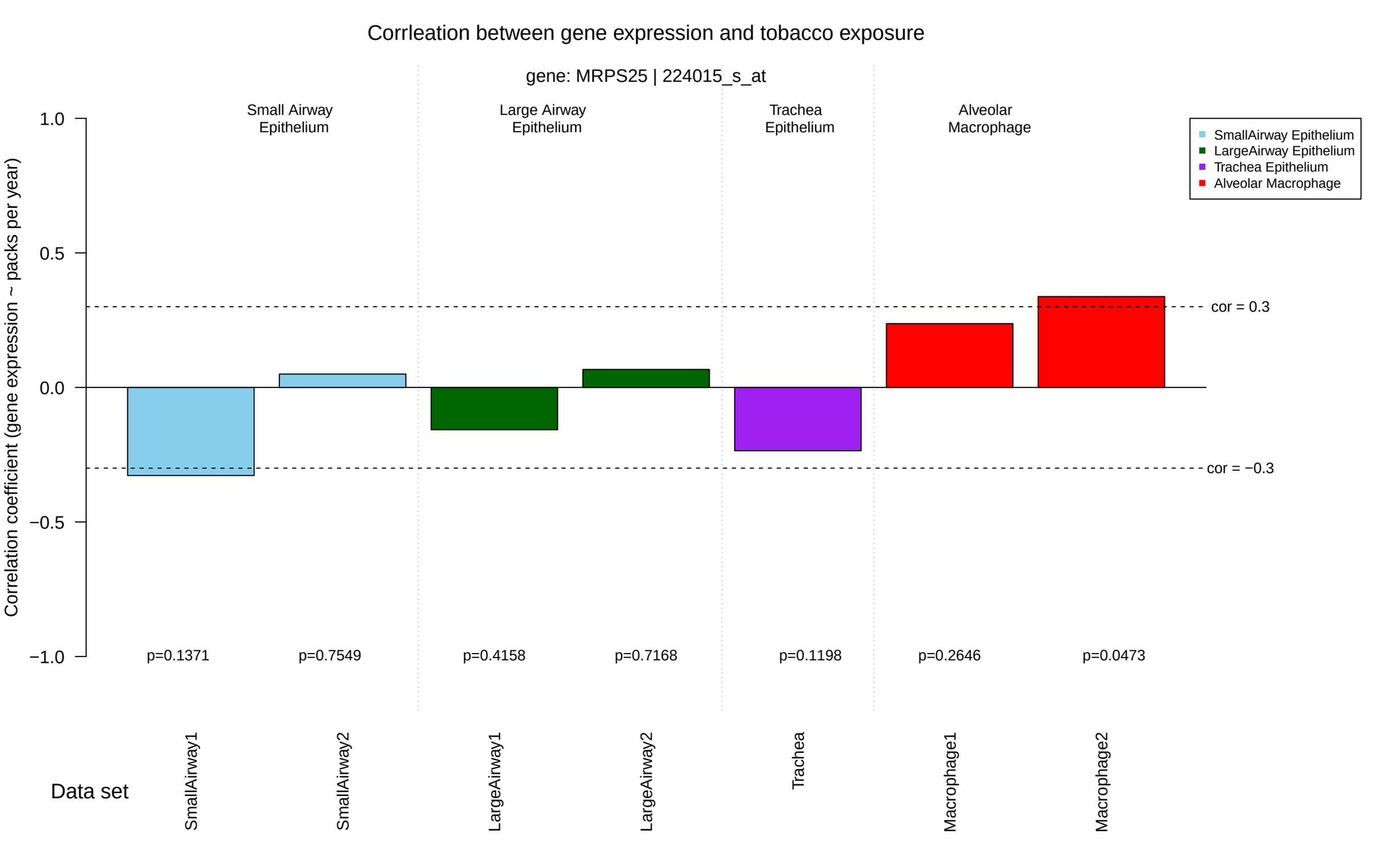

set: LargeAirway1 (p=0.0161)
MRPS25 | 224015_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4454)
MRPS25 | 224015_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2595)
MRPS25 | 224015_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.735)
MRPS25 | 224015_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0267)
MRPS25 | 224015_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MRPS25 | 224015_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1371
p=0.7549
p=0.4158
p=0.7168
p=0.1198
p=0.2646
p=0.0473
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2